

Name _____

Date March 20, 2025

Total
18
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1. Determine the critical values for $f(x) = \frac{3}{4}x^4 + \frac{10}{3}x^3 + \frac{7}{2}x^2$.

$$f(x) = \frac{3}{4}x^4 + \frac{10}{3}x^3 + \frac{7}{2}x^2$$

$$f'(x) = \frac{3}{4}(4)x^3 + \frac{10}{3}(3)x^2 + \frac{7}{2}(2)x$$

$$= 3x^3 + 10x^2 + 7x$$

$$= x(3x^2 + 10x + 7)$$

$$= x(x+1)(3x+7)$$

$$x = -1, -\frac{7}{3}, 0$$

(2)

2. Given $f'(x) = -2x(x+2)^2(x-5)$, state the intervals of increase/decrease of the original function $f(x)$ and classify all local extrema.

$$x = 0, -2, 5$$

intervals	$(-\infty, -2)$	$x = -2$	$(-2, 0)$	$x = 0$	$(0, 5)$	$x = 5$	$(5, \infty)$
$f'(x)$	$(-)$	0	$(-)$	0	$(+)$	0	$(-)$
$f(x)$	DEC	HT	DEC	MIN	INC	MAX	DEC

(4)

Intervals of Increase: $(0, 5)$

Intervals of Decrease: $(-\infty, -2) \cup (-2, 0) \cup (5, \infty)$

Local Maximum: $x = 5$

Local Minimum: $x = 0$

3. Determine the equation of the horizontal asymptote for the function $f(x) = \frac{-x^2+2x+3}{2x^2+3x+1}$. State whether the graph approaches the asymptote from above or below.

$$f(x) = \frac{-x^2+2x+3}{2x^2+3x+1}$$

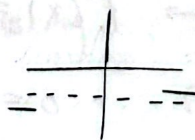
$$\lim_{x \rightarrow +\infty} f(x) = \frac{-1 + \frac{2}{x} + \frac{3}{x^2}}{2 + \frac{3}{x} + \frac{1}{x^2}}$$

$$\lim_{x \rightarrow +\infty} f(x) = -\frac{1}{2}$$

$$HA = -\frac{1}{2}$$

$$f(1000) = -\frac{1}{2}^+$$

$$f(-1000) = -\frac{1}{2}^-$$



$$\lim_{x \rightarrow +\infty} f(x) = -\frac{1}{2}^+$$

approaches asymptote from above on the right

$$\lim_{x \rightarrow -\infty} f(x) = -\frac{1}{2}^-$$

approaches asymptote from below on the left

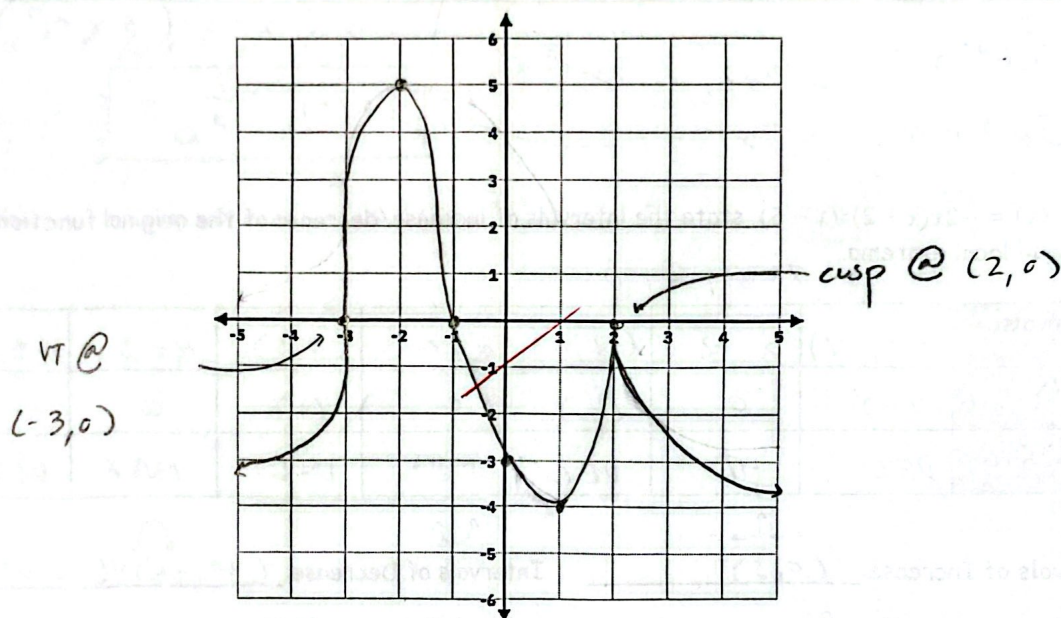
(3)

4. True or False: "When the graph of the derivative function, $f'(x)$, is decreasing, the original function must also be decreasing". Justify your answer.

FALSE. When the graph of $f(x)$ is decreasing, the graph of $f'(x)$ does not necessarily have to be decreasing, it just must be in the $(-)$ y -values (below x -axis). This is because $f'(x)$ is a measure of the slope of $f(x)$.

5. A calculus student's analysis of a function $f(x)$ results in the information below. $f(x)$ is defined for all real numbers and $f(-2) = 5$ and $f(1) = -4$. $f(x)$ has x-intercepts at -3, -1 and 2 and a y-intercept at -3. Complete the table below and sketch a possible graph that satisfies all the given conditions.

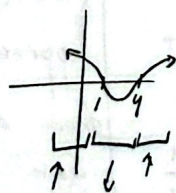
intervals	$x < -3$	$x = -3$	$-3 < x < -2$	$x = -2$	$-2 < x < 1$	$x = 1$	$1 < x < 2$	$x = 2$	$x > 2$
$f'(x)$	positive	dne	positive	0	negative	0	positive	dne	negative
$f(x)$	INC	VT	INC	max	DEC	min	INC	cusp	DEC



6. Determine the values of a, b, and c such that the cubic function $f(x) = ax^3 + bx^2 + cx$ satisfies all of the following conditions:

$$f(1) = 11; f(4) = -16; f'(x) > 0 \text{ when } x < 1 \text{ and } x > 4; f'(x) < 0 \text{ when } 1 < x < 4$$

$$f(1) = 11 \quad f(4) = -16 \quad f'(1) = 0 \quad f'(4) = 0$$



$$f(x) = ax^3 + bx^2 + cx \rightarrow f'(x) = 3ax^2 + 2bx + c$$

$$11 = a(1)^3 + b(1)^2 + c(1) \quad 0 = 3a(1)^2 + 2b(1) + c$$

$$\textcircled{1} \quad 11 = a + b + c \quad \textcircled{2} \quad 0 = 3a + 2b + c$$

$$-16 = a(4)^3 + b(4)^2 + c(4)$$

$$-16 = 64a + 16b + 4c$$

$$\textcircled{2} \quad -4 = 16a + 4b + c$$

COMPLETED ON SCRAP

$$\boxed{a = 2, \quad b = -15, \quad c = 24}$$

$$11 = a + b + c$$

$$\textcircled{2} -4 = 16a + 4b + c$$

$$\textcircled{3} 0 = 3a + 2b + c \rightarrow c = -3a - 2b$$

$$\textcircled{3} \rightarrow \textcircled{1}$$

$$\textcircled{3} \rightarrow \textcircled{2}$$

$$11 = a + b + (-3a - 2b)$$

$$-4 = 16a + 4b + (-3a - 2b)$$

$$\textcircled{4} 11 = -2a - b$$

$$\textcircled{5} -4 = 13a + 2b$$

$$b = -2a - 11$$

$$\textcircled{1} \rightarrow \textcircled{5}$$

$$-4 = 13a + 2(-2a - 11)$$

$$-4 = 13a - 4a - 22$$

$$18 = 9a$$

$$\boxed{a = 2}$$

$$-4 = 13(2) + 2b$$

$$11 = (2) + (-15) + c$$

$$-4 = 26 + 2b$$

$$\boxed{c = 24}$$

$$\boxed{b = -15}$$

5. So, if $f(x)$ is decreasing, the slope ($f'(x)$) is negative, but not always decreasing.

ex.

